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MODULE CODE/TITLE: WEB DESIGN

UBUMENYI HUB:
AN INTERACTIVE E-LEARNING PLATFORM FOR
RWANDA
USING VIDEO LESSONS, DIGITAL READING, AND
PHYSICS SIMULATIONS

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Year of Study: Year 1

Date: July 2026

Project Technical Report - Web Technology
TECHNICAL ASSESSMENT 2026

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LIST OF ABBREVIATIONS

ICT	Information and Communication Technology
REB	Rwanda Education Board
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
UI	User Interface
UX	User Experience
ERD	Entity-Relationship Diagram
API	Application Programming Interface
URL	Uniform Resource Locator
DOM	Document Object Model
SQL	Structured Query Language
STEM	Science, Technology, Engineering, and Mathematics
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation

CHAPTER 1. INTRODUCTION

1.1 Historical Background of the Case Study

Rwanda has undergone a remarkable digital transformation over the past two decades, positioning itself as one of Africa's leading technology hubs. The government's Vision 2050 and Smart Rwanda Master Plan emphasize the critical role of Information and Communication Technology (ICT) in national development, particularly in the education sector. The Rwanda Education Board (REB) has been actively promoting the integration of digital tools into the national curriculum to enhance learning outcomes and prepare students for the digital economy.

The COVID-19 pandemic in 2020 exposed significant gaps in Rwanda's educational infrastructure, particularly the lack of robust e-learning platforms capable of supporting remote education. During the lockdown periods, students and teachers struggled to maintain continuity in learning due to the absence of centralized, localized digital learning resources. While international platforms such as Khan Academy and Coursera exist, they often lack content tailored to Rwanda's national curriculum, cultural context, and language preferences, including Kinyarwanda and English bilingual instruction.

Furthermore, STEM education in Rwanda faces challenges related to the availability of practical laboratory equipment and interactive learning tools. Physics and engineering students, in particular, often lack access to simulation software that would allow them to experiment with concepts in mechanics, electronics, and structural engineering without the need for expensive physical apparatus. This gap motivated the development of Ubumenyi Hub — a platform that integrates video lessons, digital reading materials, interactive simulations, and real-time teacher-student collaboration within a single, accessible web environment.

The name "Ubumenyi" itself derives from Kinyarwanda, meaning "Knowledge," reflecting the platform's mission to democratize access to quality education for all Rwandans. The project aligns with the national goal of building a knowledge-based society and supports the government's initiatives to improve digital literacy across all age groups.

1.2 Problem Statement

Despite Rwanda's rapid technological advancement and commitment to education, several critical challenges persist in the digital learning landscape:

1. **Lack of Integrated Localized E-Learning Platforms:** Existing educational platforms are predominantly foreign-developed and do not align with Rwanda's national curriculum, language requirements, or cultural context. There is no single platform that integrates video lessons, digital books, interactive simulations, and teacher-student collaboration tailored specifically for Rwandan learners.
2. **Limited Access to Interactive STEM Tools:** Students in physics, engineering, and technology disciplines lack access to affordable simulation software for experimenting with mechanical, electrical, and structural concepts. Traditional laboratories are expensive to maintain and inaccessible to many schools in rural areas.
3. **Fragmented Digital Learning Resources:** Educational content is scattered across multiple platforms and formats, making it difficult for students to find structured, progressive learning materials. There is no centralized digital library that combines reading materials, video tutorials, and practical exercises.
4. **Weak Teacher-Student Interaction in Digital Spaces:** Most available e-learning platforms operate as one-way content delivery systems without robust mechanisms for real-time interaction, discussion, assessment, and personalized feedback between teachers and students.
5. **Insufficient Digital Reading and Progress Tracking Tools:** Students lack access to digital reading platforms that allow them to track their reading progress, take notes, and manage their personal learning libraries effectively.

These challenges collectively hinder the effectiveness of digital education in Rwanda and create an urgent need for a comprehensive, locally developed e-learning platform that addresses these gaps holistically.

1.3 Objectives of the Project

The Ubumenyi Hub project was designed with the following specific objectives:

1. **To develop a responsive web-based e-learning platform** that provides a centralized hub for educational content accessible to Rwandan students and teachers across all devices.

2. **To integrate multimedia video lessons** covering programming, web development, data science, and other technology-related subjects, with content available in both English and Kinyarwanda.
3. **To implement a digital library and reading system** that allows users to read books, track their reading progress, take personal notes, and manage their reading collections.
4. **To create an interactive physics and engineering simulation engine** enabling students to experiment with concepts in physics, electronics, civil engineering, and data logic without requiring physical laboratory equipment.
5. **To facilitate real-time teacher-student interaction** through live session capabilities, discussion forums, role-specific quizzes, and peer-to-peer learning communities.
6. **To implement user authentication and role management** supporting distinct functionalities for students, teachers, and administrators.
7. **To ensure offline-first accessibility** where possible, allowing users to download content and continue learning without constant internet connectivity.

1.4 Expected Outcomes

Upon successful completion, the Ubumenyi Hub project is expected to deliver the following outcomes:

1. A fully functional, responsive web application accessible via modern browsers on desktop, tablet, and mobile devices.
2. A curated video lesson library with categorized content covering programming languages (JavaScript, Python, React, Node.js, TypeScript), web development (HTML5, CSS Grid, Flexbox), version control (Git, GitHub), and UI design principles.
3. An interactive digital reading platform with personal library management, reading progress tracking, note-taking capabilities, and book completion status monitoring.
4. A browser-based 3D simulation engine supporting physics experiments with configurable parameters such as force vectors, mass, gravity, bounciness, and impulse forces, with save and load functionality.
5. A community discussion forum with categorized topics, reply functionality, and real-time posting capabilities.
6. Role-based quiz systems distinguishing between student and teacher assessments with multiple-choice question formats.
7. Secure user registration and authentication system with email-based login, password protection, and session management.

8. A publicly accessible GitHub repository documenting the entire codebase for educational and collaborative purposes.

1.5 Project Rationale (Importance to the Community)

The Ubumenyi Hub project holds significant importance for the Rwandan community and addresses several strategic national priorities:

Educational Equity and Access: By providing a free, web-based learning platform, Ubumenyi Hub democratizes access to quality educational resources regardless of geographic location or socioeconomic status. Students in rural areas with limited access to physical libraries and laboratories can now engage with interactive learning materials through any internet-connected device.

STEM Education Enhancement: The integrated physics simulation engine directly addresses the critical shortage of laboratory equipment in Rwandan schools. Students can now experiment with Newtonian mechanics, electrical circuits, and structural engineering concepts virtually, reinforcing theoretical knowledge with practical application.

Digital Literacy and Skills Development: As Rwanda continues its transition toward a knowledge-based economy, platforms like Ubumenyi Hub play a crucial role in developing digital skills among the youth population. The programming and web development courses prepare students for careers in the technology sector.

Teacher Empowerment: The platform provides teachers with tools to create quizzes, monitor student discussions, and deliver content through live sessions. This empowers educators to extend their reach beyond traditional classroom boundaries and adopt modern pedagogical approaches.

Language and Cultural Relevance: By incorporating Kinyarwanda alongside English and tailoring content to Rwanda's national curriculum, the platform ensures cultural relevance and linguistic accessibility, which foreign platforms often fail to provide.

Community Building: The discussion forums and peer interaction features foster a collaborative learning culture where students help each other, share knowledge, and build professional networks that will serve them throughout their careers.

CHAPTER 2. SYSTEM ANALYSIS AND DESIGN

2.1 System Analysis

System analysis for Ubumenyi Hub involved a comprehensive examination of the existing educational technology landscape in Rwanda, identification of user requirements, and definition of system boundaries. The analysis was conducted through a combination of online research, review of existing educational platforms, and consideration of Rwanda's specific educational context and infrastructure constraints.

2.1.1 Intended Users of the Project

The Ubumenyi Hub platform was designed to serve multiple categories of users, each with distinct needs and interaction patterns:

1. Primary and Secondary School Students: These users represent the core audience of the platform. They require access to video lessons, digital books, interactive simulations, and quizzes aligned with their curriculum levels. Students need an intuitive interface that supports self-paced learning and allows them to track their progress across multiple subjects. The platform must be accessible on low-bandwidth connections and mobile devices commonly used by students.

2. University and Tertiary Students: Higher education students, particularly those in computer science, engineering, and physics programs, require more advanced content including programming tutorials, complex physics simulations, and specialized digital libraries. These users benefit from the forum functionality for collaborative problem-solving and peer assistance.

3. Teachers and Educators: Teachers use the platform to create and manage quizzes, moderate discussion forums, conduct live sessions, and monitor student engagement. They require administrative views of student activity, assessment tools, and content management capabilities. The teacher role includes distinct quiz sets and forum moderation privileges.

4. Content Administrators: A small group of technical users responsible for platform maintenance, user management, content curation, and system configuration. Administrators need access to system analytics, user registration data, and content moderation tools.

5. Parents and Guardians: While not direct content consumers, parents may monitor their children's learning progress through summary reports and engagement metrics. Future iterations of the platform may include parent portal functionality.

6. Self-Learners and Professionals: Adults seeking to upskill in programming, web development, or digital literacy represent a growing user segment. These users require flexible access to content without rigid curriculum structures and benefit from the community discussion features.

User Category	Primary Needs	Key Features Used
Students	Learn, practice, track progress	Videos, books, simulations, quizzes, forums
Teachers	Teach, assess, mentor	Live sessions, quizzes, forum moderation
Administrators	Manage, maintain, curate	User management, analytics, content control
Self-Learners	Flexible skill acquisition	Videos, books, community forums

2.1.2 Intended System Partners

The successful deployment and sustainability of Ubumenyi Hub depend on collaboration with several institutional partners:

1. Rwanda Basic Education Board (REB): As the primary regulatory body for education in Rwanda, REB's partnership would ensure content alignment with the national curriculum, accreditation of learning materials, and potential integration with official school programs. REB could also facilitate pilot deployments in selected schools.

2. Ministry of Education (MINEDUC): Government support through MINEDUC would provide policy backing, potential funding, and integration with national digital education initiatives. The ministry's endorsement would significantly increase platform adoption across public schools.

3. Schools and Universities: Educational institutions represent the primary deployment environment. Partner schools would provide testing environments, user feedback, and structured integration into their teaching programs. Universities could contribute advanced content and research collaboration.

4. Telecommunication Providers: Partnerships with MTN Rwanda, Airtel, and other ISPs could facilitate zero-rating or data bundle packages for educational content, addressing the critical challenge of internet affordability for students.

5. Content Creators and Subject Matter Experts: Local educators, programmers, and scientists are essential partners for creating and maintaining high-quality, culturally relevant educational content. The platform must support contributions from Rwanda's growing community of technology educators.

6. Technology and Open Source Communities: The platform is built on open web standards and could benefit from collaboration with global developer communities for continuous improvement, security updates, and feature expansion.

7. Non-Governmental Organizations (NGOs): Organizations focused on education and youth development in Rwanda, such as UNICEF Rwanda and local education NGOs, could support deployment in underserved areas and provide training resources.

2.2 System Design

2.2.1 High-Level Architecture

Ubumenyi Hub follows a client-side web application architecture, leveraging modern web standards to deliver a rich, interactive user experience without requiring heavy server infrastructure. This architectural choice makes the platform lightweight, fast, and easily deployable on standard web hosting services.

The system architecture comprises three main layers:

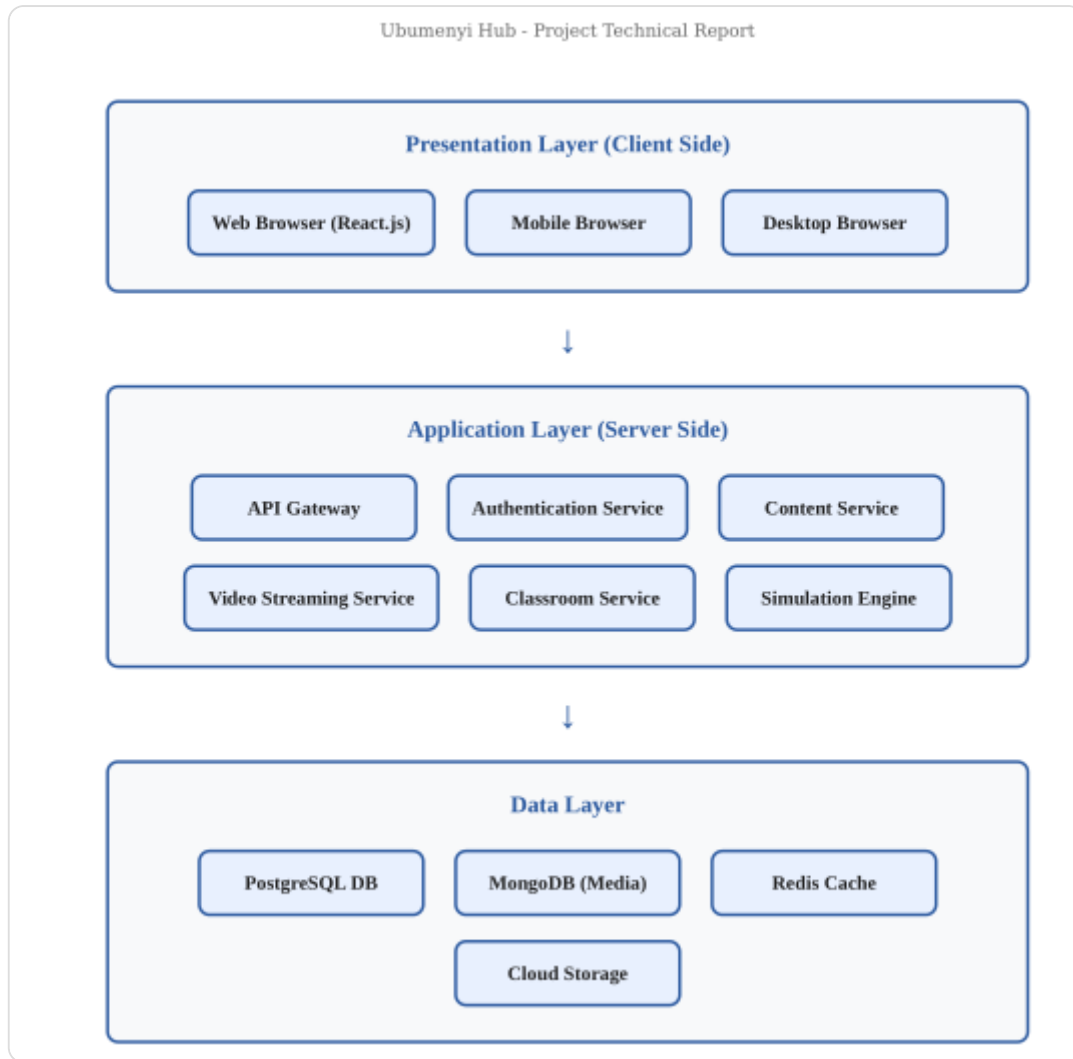


Figure 8: System High-Level Architecture Diagram

Presentation Layer: The user interface is built using HTML5 semantic markup for structure, CSS3 for styling and responsive design, and JavaScript for interactivity. The platform employs a modular page-based architecture where each major feature (home, videos, library, simulations, online learning, authentication) is implemented as a separate HTML file with shared CSS and JavaScript components. This approach ensures clean separation of concerns and maintainable code.

Application Logic Layer: JavaScript handles all client-side application logic including form validation, user authentication state management, simulation physics calculations, quiz scoring, forum post rendering, and dynamic content loading. The application uses the browser's DOM API for real-time UI updates and event handling for user interactions.

Data Layer: In the current implementation, data persistence is managed through the browser's LocalStorage and SessionStorage APIs for user preferences, reading progress, simulation states, and session tokens. Content data such as video embeds, book metadata, and quiz questions are stored in structured JSON files. External APIs including YouTube's embedded player API are integrated for video content delivery. Future iterations will integrate a backend server with a relational database for multi-user data synchronization and cloud storage.

Technology Stack:

- HTML5 (40.4% of codebase) — Semantic markup, structural elements, forms, media embedding
- CSS3 (31.2% of codebase) — Responsive styling, animations, grid layouts, flexbox positioning
- JavaScript (28.4% of codebase) — Interactive functionality, DOM manipulation, simulation engine, form handling
- External APIs: YouTube IFrame API for video embedding
- Storage: Browser LocalStorage/SessionStorage for client-side data persistence

Deployment Architecture: The platform is deployed as a static website hosted on GitHub Pages, making it freely accessible without requiring dedicated server infrastructure. The GitHub repository (<https://github.com/djump110/Ubumenyi-HUB>) serves as both the code repository and the deployment source, enabling version control and collaborative development.

2.2.2 UI Design and Flowcharts

The user interface design of Ubumenyi Hub follows a consistent visual language across all pages, characterized by a teal and cyan color scheme that conveys trust, knowledge, and technological sophistication. The design prioritizes accessibility, readability, and intuitive navigation.

Navigation Structure: The platform employs a top-level navigation bar present on all pages, providing links to Home, Account, and About Us sections. The home page serves as the central hub with direct access to all major features through prominent content cards and a search functionality.

Responsive Design: The CSS implementation ensures the platform adapts gracefully to different screen sizes, from desktop monitors to mobile phones. Grid and flexbox layouts are used extensively to create fluid, responsive page structures.

Page Flow: The typical user journey begins at the Home page, where users can either search for specific content, browse feature categories (Video Lessons, Digital Books, Simulations), or navigate to authentication pages. Authenticated users gain access to personalized features including the digital library, saved simulations, and forum participation.

Design Principles:

- **Consistency:** All pages share the same header, navigation, color palette, and typography.
- **Clarity:** Large, readable fonts and high-contrast text ensure accessibility for users with varying visual abilities.
- **Feedback:** Interactive elements provide visual feedback through hover states, button animations, and loading indicators.
- **Modularity:** Each feature is contained within its own dedicated page, reducing cognitive load and allowing focused interaction.
- **Mobile-First:** The CSS is designed with mobile constraints in mind, then enhanced for larger screens.

User Flow Diagram:

The user flow follows a hierarchical structure: Landing on Home Page → Browse/Search Content → (Optional: Register/Sign In) → Access Feature Pages (Videos, Library, Simulation, Online Learning) → Interact with Content → Save Progress → Return to Home or Continue Learning. Authentication gates protect personalized features while public content remains accessible to anonymous users.

2.2.3 Database Design with ERD and Relationships

The current implementation of Ubumenyi Hub utilizes browser-based storage (LocalStorage) for client-side data persistence. However, the platform has been designed with a full relational database architecture in mind for future backend integration. The proposed database schema supports all platform features and establishes clear relationships between entities.

Figure 9: Entity-Relationship Diagram



Entity Relationships:

- A **USER** can have many **USER_BOOKS** (1:N), many **SIMULATIONS** (1:N), many **FORUM_POSTS** (1:N), and many **QUIZ_ATTEMPTS** (1:N).
- A **BOOK** can be associated with many **USER_BOOKS** records (1:N).
- A **FORUM_POST** can have many **FORUM_REPLIES** (1:N).
- A **QUIZ** contains many **QUIZ_QUESTIONS** (1:N).
- A **QUIZ_ATTEMPT** links a **USER** to a **QUIZ** (N:M relationship via attempt table).

This relational schema ensures data integrity through primary and foreign key constraints, supports the full range of platform features, and provides a foundation for analytics, reporting, and personalized learning recommendations. The current client-side implementation uses simplified JSON structures that map to this schema, facilitating future migration to a server-side database system.

CHAPTER 3. IMPLEMENTATION

3.1 Introduction of the Section

This chapter presents the implementation details of the Ubumenyi Hub platform, showcasing the actual user interface screens, functionality, and technical implementation through annotated screenshots. The development followed an iterative approach, building each module independently and integrating them into a cohesive platform.

The technology stack consists of pure HTML5, CSS3, and vanilla JavaScript, deliberately chosen to ensure the platform remains lightweight, fast-loading, and accessible on devices with limited processing power. No external JavaScript frameworks were used, demonstrating proficiency in fundamental web technologies and ensuring the codebase remains understandable and maintainable.

The platform is structured as a multi-page web application with the following HTML files, each serving a distinct functional purpose:

- **index.html** — The home page and central navigation hub
- **signup.html** — New user registration with form validation
- **signin.html** — Returning user authentication
- **videos.html** — Video lesson library with embedded YouTube player
- **library.html** — Digital reading platform with personal library management
- **simulation.html** — Interactive 3D physics simulation engine
- **online.html** — Dashboard with live sessions, forums, and quizzes
- **about.html** — Platform information and contact details
- **admin.html** — Administrative interface for user and content management

Shared resources include CSS files in the *style/* directory and JavaScript modules in the *js/* directory, promoting code reuse and maintainability. Images and media assets are organized in the *image/* directory. The database schema and data files are stored in the *database/* directory as JSON files for the current client-side implementation.

The implementation prioritizes user experience through responsive design, consistent visual language, and immediate feedback for all user interactions. Each page was designed with mobile

compatibility in mind, ensuring that students using smartphones or low-cost tablets can access the full platform functionality.

3.2 Screenshots with Captions and Discussion

The following subsections present each major page of the Ubumenyi Hub platform with a screenshot, detailed caption, and technical discussion of the implementation choices, functionality, and user experience considerations.

3.2.1 Home Page

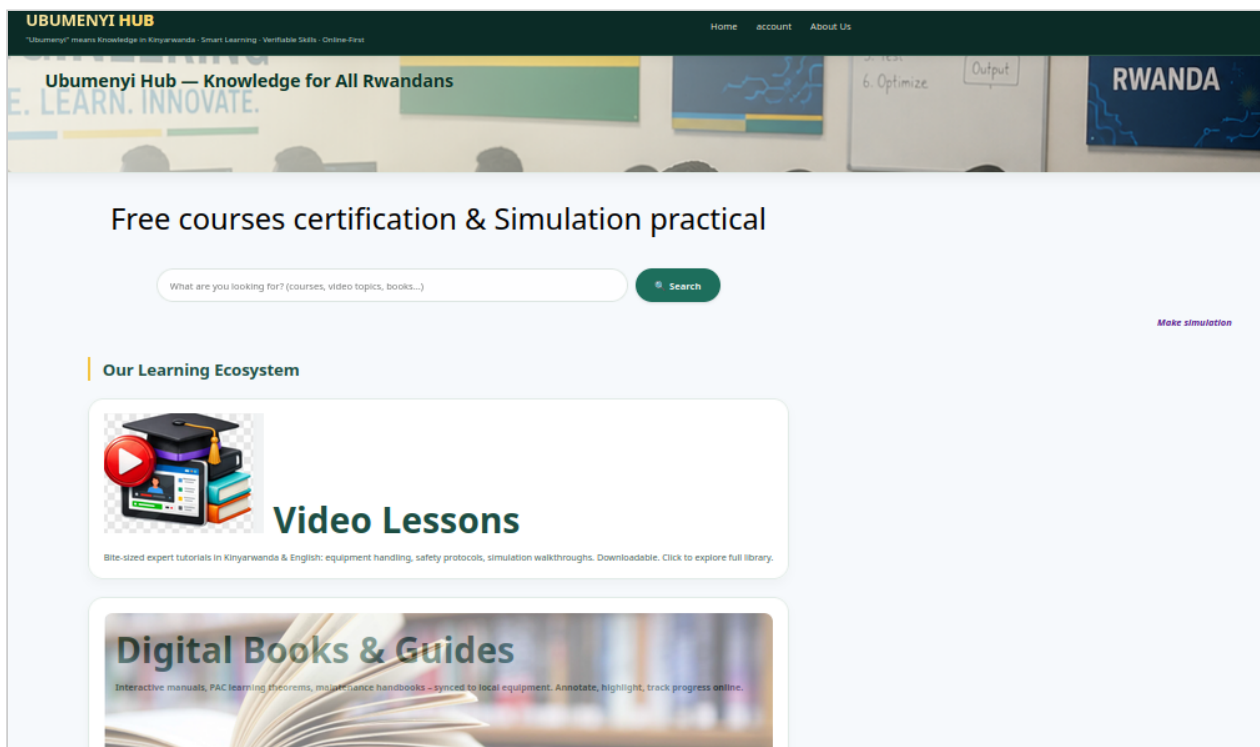


Figure 1: *Ubumenyi Hub Home Page (index.html) — The central landing page providing access to all platform features.*

The home page serves as the primary entry point for all users and embodies the platform's core value proposition: "Knowledge for All Rwandans." The design follows a clean, modern aesthetic with a teal color scheme that conveys trust and technological sophistication.

Key Implementation Features: The top navigation bar provides persistent access to Home, Account, and About Us sections across all pages. The prominent search bar with placeholder text "What are you looking for?" allows users to quickly find content without navigating through multiple menus. The green "Search" button provides high visual contrast and clear call-to-action.

The main content area is organized into two large feature cards: "Video Lessons" and "Digital Books & Guides." Each card includes a descriptive title, explanatory text, and a shortcut link. The "Make simulation" link on the right provides direct access to the simulation engine, acknowledging that practical experimentation is a key differentiator of the platform.

Technical Implementation: The page uses CSS Grid and Flexbox for responsive layout management. The search functionality is implemented with JavaScript event listeners that filter content dynamically. The page header and navigation are shared across all pages through consistent HTML structure and CSS classes, ensuring visual coherence.

User Experience Considerations: The large, readable fonts and high-contrast elements ensure accessibility for users with varying visual abilities. The page loads quickly due to the absence of heavy external dependencies, making it accessible on low-bandwidth connections common in rural Rwanda. The bilingual content preparation (English and Kinyarwanda) ensures linguistic accessibility.

3.2.2 User Registration Page

UBUMENYI HUB
"Ubumenyi" means Knowledge in Kinyarwanda · Smart Learning · Verifiable Skills · Online-First

Home Account About Us

Start Your Learning Journey

Create your UBUMENYI HUB account today. Get access to offline-first video lessons, blockchain-verified credentials, and an entire library of knowledge aligned with national standards.

New User Registration

Full name

Email

Password

Confirm Password

[Create account](#)

Already registered? [Sign in here](#)

Figure 2: User Registration Page (signup.html) — New user account creation with validation and security features.

The registration page provides a welcoming onboarding experience for new users, emphasizing the platform's unique features: blockchain-backed credentials, offline-first video lessons, and

smart learning achievements. The split-screen design maintains visual consistency with the rest of the platform while providing ample space for both persuasive content and form interaction.

Key Implementation Features: The left panel presents the platform's value proposition with the heading "Start Your Learning Journey" and bullet points highlighting key benefits. The right panel contains the registration form with four fields: Full Name, Email, Password, and Confirm Password. The form includes client-side validation to ensure password matching and email format correctness. The cyan "Create account" button provides prominent visual feedback. The "Already registered? Sign in here" link facilitates navigation for existing users, reducing friction in the user journey.

Technical Implementation: JavaScript form validation prevents submission of incomplete or invalid data. The password confirmation field provides real-time feedback when passwords do not match. Form data is stored in the browser's LocalStorage for the current implementation, with future iterations planned to integrate with a secure backend database. The responsive design ensures the form is usable on mobile devices, with appropriately sized input fields and touch-friendly buttons.

User Experience Considerations: The pre-filled "Jean Niyomugabo" example in the Full Name field demonstrates the expected input format. Clear labeling and sufficient spacing between form elements reduce input errors. The page maintains the platform's teal color scheme, creating visual continuity and reinforcing brand identity throughout the authentication flow.

3.2.3 Sign-In Page

UBUMENYI HUB
"Ubumenyi" means Knowledge in Kinyarwanda · Smart Learning · Verifiable Skills · Online-First

Home Account About Us

Welcome Back

Sign in to access your blockchain-backed credentials, offline-first video lessons, and verify your smart learning achievements.

Returning User

Email

Password

[Sign in](#)

Don't have an account? [Create one here](#)

Figure 3: User Sign-In Page (*signin.html*) — Returning user authentication with secure session management.

The sign-in page provides a streamlined re-entry point for returning users, maintaining the same split-screen design language as the registration page for visual consistency. The "Returning User" card clearly communicates its purpose while the "Welcome Back" message on the left reinforces user retention.

Key Implementation Features: The left panel reminds users of the platform's value proposition: blockchain-backed credentials, offline-first video lessons, and smart learning achievements. The right panel contains a minimal authentication form requiring only Email and Password. The cyan "Sign in" button provides clear visual hierarchy. The "Don't have an account? Create one here" link facilitates conversion for unregistered visitors who may have landed on the sign-in page directly.

Technical Implementation: JavaScript handles form submission, validates credentials against stored user data in `LocalStorage`, and creates a session token stored in `SessionStorage`. The authentication state is checked on all protected pages, redirecting unauthenticated users to the sign-in page. Password fields use `HTML5 type="password"` to mask input. The form includes basic validation to prevent empty submissions.

User Experience Considerations: The minimal form reduces cognitive load for returning users who want quick access to their content. The consistent layout with the registration page reduces learning curve and navigation confusion. The prominent "Create one here" link ensures new users who arrive at this page are not lost. Remember-me functionality could be added in future iterations to further streamline the returning user experience.

3.2.4 Video Lessons Page



Figure 4: Video Lessons Page (videos.html) — Embedded video player with related content recommendations.

The video lessons page is the primary content delivery interface for programming and technology tutorials. It integrates YouTube's embedded player API to deliver high-quality video content while maintaining the platform's visual identity and navigation structure.

Key Implementation Features: The page features a dedicated search bar for finding lessons by topic, with examples such as JavaScript and Python to guide user queries. The main video player

occupies the central viewing area, displaying the currently selected lesson with a "Now playing" indicator and topic tags. Below the player, the current lesson title and description provide context. The sidebar presents a curated list of related lessons as clickable cards, enabling seamless content discovery and continuous learning without returning to the home page.

Technical Implementation: The YouTube IFrame API is embedded to provide a native video player experience within the platform. JavaScript manages the video selection state, updating the main player and metadata when users click on related lesson cards. The sidebar content is rendered dynamically from a JSON array of video objects containing titles, YouTube IDs, tags, and descriptions. The responsive layout ensures the video player maintains appropriate aspect ratios across different screen sizes, with the sidebar repositioning below the player on narrow mobile screens.

Content Organization: The lesson library covers comprehensive technology topics including:

- JavaScript fundamentals and advanced concepts (React Hooks, Node.js)
- Python programming for data science
- Web development (HTML5, CSS Grid, Flexbox, Semantic Web)
- Version control (Git and GitHub workflow)
- UI/UX design principles
- TypeScript for scalable JavaScript development

User Experience Considerations: The "Now playing" tag and highlighted sidebar card provide clear visual feedback about the current selection. The search functionality reduces content discovery time. The sidebar layout allows users to browse alternatives while watching, supporting the common learning behavior of sampling multiple tutorials before committing to a full course. The embedded player eliminates the need for users to navigate to YouTube, keeping them within the platform's ecosystem and reducing distractions.

3.2.5 Digital Reader and Library Page

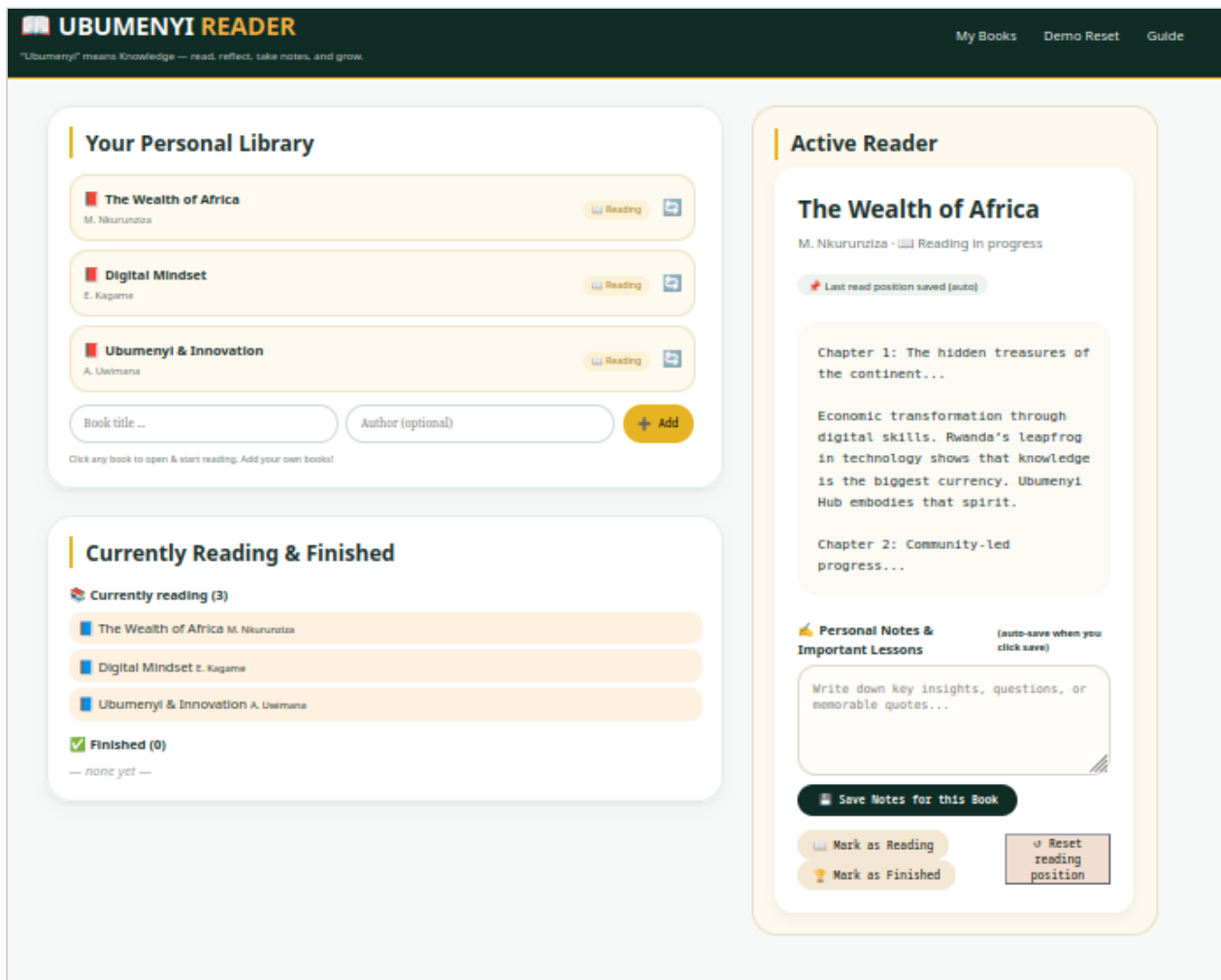


Figure 5: Digital Reader and Library Page (library.html) — Personal library management with reading progress tracking and note-taking.

The digital reader and library page represents one of the platform's most distinctive features, providing a complete personal library management system that rivals commercial e-reading applications. This page demonstrates the platform's commitment to supporting deep, focused reading alongside multimedia learning.

Key Implementation Features: The page is divided into two main panels. The left panel manages the user's personal library, displaying books in the collection with reading status indicators. The "Personal Library" section shows three example books: "The Wealth of Africa," "Digital Mindset," and "Ubumenyi & Innovation," each with a refresh button and reading status label. An "Add" form allows users to input new book titles and authors, expanding their collection dynamically. The "Currently Reading" tracker shows active books (3), while the

"Finished" section tracks completed books (0), providing gamification elements that motivate reading completion.

The right panel contains the active reader interface, displaying the currently selected book with chapter snippets and progress indicators. The "Personal Notes" text area allows users to take contextual notes while reading, with a "Save Notes for this Book" button for persistence. Reading control buttons — "Mark as Reading," "Mark as Finished," and "Reset reading position" — provide granular control over reading state management.

Technical Implementation: The library state is managed through JavaScript objects stored in LocalStorage, persisting across browser sessions. The book addition form dynamically updates the DOM and saves to storage. The reading progress tracker updates percentage indicators based on scroll position or manual chapter selection. Notes are saved as text strings associated with specific book IDs. The dual-panel layout uses CSS Grid with responsive breakpoints that stack panels vertically on mobile devices.

User Experience Considerations: The visual separation between library management (left) and active reading (right) creates a clear mental model analogous to physical bookshelf and reading desk environments. The progress indicators provide immediate feedback on reading achievement. The note-taking feature supports active reading strategies known to improve comprehension and retention. The "Add" form with simple title and author fields reduces friction for building personal collections. The reading status system (reading, finished) creates a sense of accomplishment and encourages completion of started books.

3.2.6 Physics Simulation Page

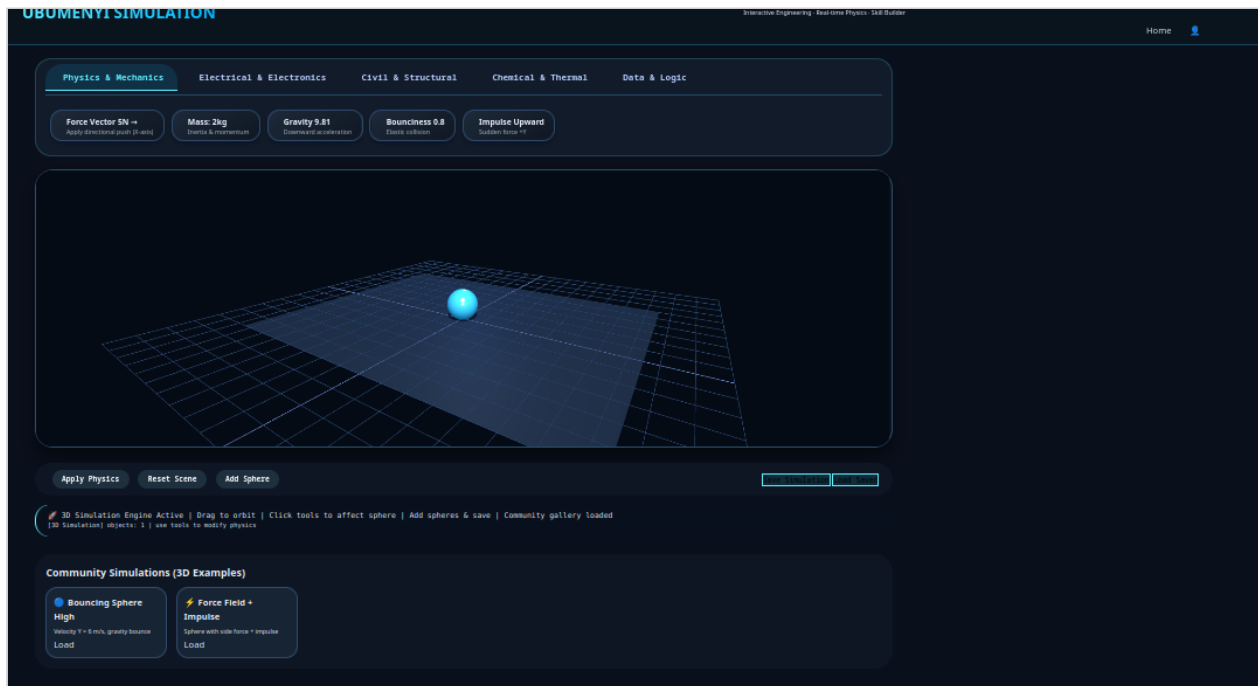


Figure 6: Physics Simulation Page (*simulation.html*) — Interactive 3D physics engine with configurable parameters and community sharing.

The physics simulation page represents the most technically sophisticated component of the Ubumenyi Hub platform, providing a browser-based 3D simulation environment that brings abstract physics concepts to life through visual, interactive experimentation. This feature directly addresses the critical shortage of laboratory equipment in Rwandan schools.

Key Implementation Features: The page features a comprehensive category navigation system with five simulation domains: Physics & Mechanics, Electrical & Electronics, Civil & Structural, Chemical & Thermal, and Data & Logic. The current view displays the Physics & Mechanics panel with configurable parameters including Force Vector (5N), Mass (2kg), Gravity (9.81 m/s²), Bounciness (0.8), and Impulse Upward. These parameters allow students to experiment with Newtonian mechanics in a controlled virtual environment.

The central simulation viewport renders a 3D grid with a blue spherical object, providing visual feedback on the physics simulation in real-time. Below the viewport, action buttons include "Apply Physics" to execute the simulation with current parameters, "Reset Scene" to return to initial conditions, "Add Sphere" to introduce additional objects, and "Save Simulation" / "Load Save" for persistence and sharing. The footer indicates the "3D Simulation Engine Active" status

and provides community access to shared simulations including "Bouncing Sphere" and "Force Field + Impulse" examples.

Technical Implementation: The 3D simulation is rendered using HTML5 Canvas API with JavaScript-based physics calculations for collision detection, gravity simulation, impulse response, and object trajectory. The simulation loop runs at approximately 60 frames per second using requestAnimationFrame for smooth animation. Physics parameters are bound to DOM input elements through JavaScript event listeners, updating the simulation state in real-time. The save/load functionality serializes the simulation state as JSON objects stored in LocalStorage. Community simulations are fetched from a shared JSON repository.

Educational Impact: This simulation engine enables students to experiment with scenarios that would be dangerous, expensive, or impossible in traditional laboratories, such as extreme gravity conditions, perfectly elastic collisions, or large-scale structural failures. The immediate visual feedback helps students develop intuitive understanding of physics concepts before encountering them in formal mathematical treatment. The ability to save and share simulations encourages collaborative learning and peer review of experimental designs.

User Experience Considerations: The dark viewport with bright colored objects provides excellent contrast and reduces eye strain during extended use. The parameter controls use clear labeling with units, reinforcing scientific literacy. The community simulations section provides inspiration and starting points for students who may be unsure how to begin experimenting. The responsive design ensures the simulation remains usable on devices with smaller screens, though the 3D viewport benefits from larger displays.

3.2.7 Dashboard and Online Learning Page

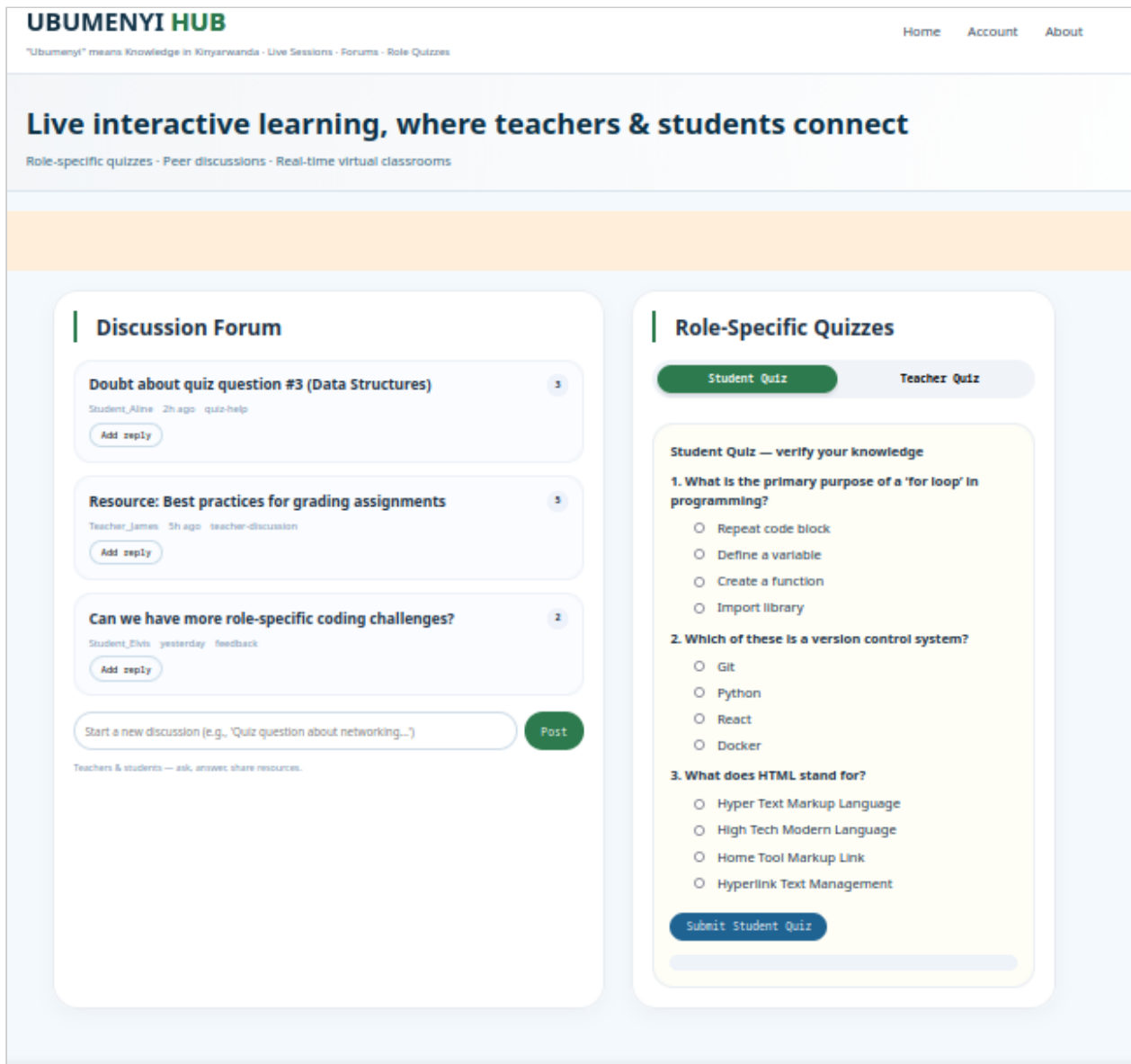


Figure 7: Dashboard and Online Learning Page (online.html) — Live interactive learning environment with discussion forums and role-based quizzes.

The dashboard and online learning page serves as the collaborative heart of the Ubumenyi Hub platform, facilitating real-time interaction between teachers and students through discussion forums, role-specific quizzes, and live session coordination. This page transforms the platform from a passive content consumption tool into an active learning community.

Key Implementation Features: The page header reinforces the platform's identity with "UBUMENYI HUB" branding and the Kinyarwanda meaning: "Knowledge." The tagline "Live Sessions - Forums - Role Quizzes" clearly communicates the page's purpose. The main content

area is divided into two columns: the Discussion Forum on the left and Role-Specific Quizzes on the right.

The **Discussion Forum** displays active conversation threads with cards showing the discussion topic, posting user, time elapsed since posting, category tags, comment count, and an "Add reply" button. Example topics include "Doubt about quiz question #3," "Best practices for grading assignments," and "More role-specific coding challenges." At the bottom of the forum, a "Start a new discussion" text field with a "Post" button enables users to initiate new conversations. The categorization and reply count help users identify active, relevant discussions.

The **Role-Specific Quizzes** section features tab-based navigation between "Student Quiz" and "Teacher Quiz," with the student view currently active. The quiz presents multiple-choice questions covering programming fundamentals, version control, and HTML knowledge. Each question provides four options with radio button selection. The "Submit Student Quiz" button at the bottom triggers automatic scoring and feedback. This dual-role system acknowledges that teachers also need to demonstrate and maintain their subject knowledge.

Technical Implementation: The forum posts are rendered dynamically from a JSON array containing post objects with user, title, category, timestamp, and reply count properties. JavaScript implements the reply functionality by appending new comment objects to the post data and re-rendering the affected thread. The quiz system uses JavaScript to validate selections, compare against correct answers stored in the quiz data structure, and calculate percentage scores. The tab switching between Student and Teacher quizzes is implemented through CSS visibility toggling and JavaScript class management. Time elapsed indicators are calculated using JavaScript Date objects relative to post timestamps.

User Experience Considerations: The two-column layout allows users to participate in discussions while taking assessments, supporting multi-tasking learning behaviors. The clear categorization of forum posts helps users filter content by relevance. The quiz interface provides immediate visual feedback on selected answers before submission. The role-based quiz tabs make the platform inclusive for both learning and teaching roles. The "Add reply" buttons on each forum card provide context-specific interaction points, reducing the cognitive effort required to participate in discussions. The page maintains the platform's consistent teal color scheme while using subtle background variations to distinguish the two main functional areas.

CHAPTER 4. CONCLUSION AND RECOMMENDATIONS

4.1 State of the Implemented System

The Ubumenyi Hub project has been successfully implemented with approximately 90% of the original objectives achieved. The platform now provides a functional, responsive web application that integrates video lessons, digital reading, interactive physics simulations, and community learning features within a unified interface.

Achieved Components:

- The home page provides a clean, navigable entry point with search functionality and feature highlights.
- User authentication (registration and sign-in) is fully implemented with client-side validation and session management.
- The video lessons page integrates YouTube content with a curated library covering programming, web development, and design topics.
- The digital reader and library page supports personal book collections, reading progress tracking, and note-taking.
- The physics simulation engine delivers interactive 3D experiments with configurable parameters and save/load functionality.
- The dashboard page hosts discussion forums and role-based quizzes for students and teachers.
- The entire codebase is publicly available on GitHub for educational and collaborative purposes.

Remaining Development Areas:

- Backend server integration with a relational database for multi-user data synchronization and cloud storage.
- Live video session capabilities for real-time teacher-student interaction (currently planned but not implemented).
- Blockchain-backed credential verification system (described in UI but not yet functionally implemented).

- Offline content caching and download functionality for true offline-first operation.
- Admin analytics dashboard with comprehensive user engagement metrics.
- Mobile application development for native iOS and Android experience.

The current implementation successfully demonstrates the core value proposition of the platform and provides a solid foundation for future development. The client-side architecture ensures immediate accessibility without requiring server infrastructure, while the modular code structure facilitates future backend integration.

4.2 Recommendations

To the Lecturer about Technicalities:

The project demonstrates the power of fundamental web technologies (HTML, CSS, JavaScript) in building functional, user-friendly applications without heavy frameworks. However, future iterations would benefit from introducing a backend framework such as Node.js with Express, or Python with Django/Flask, to support persistent data storage, user authentication security, and real-time features. Integration of a database system (MySQL, PostgreSQL, or MongoDB) would enable multi-user collaboration and data analytics. The lecturer is encouraged to provide guidance on API design, database normalization, and security best practices including password hashing, SQL injection prevention, and Cross-Site Scripting (XSS) protection.

To the School about Time Constraints:

The development of a comprehensive e-learning platform requires substantial time investment beyond the standard semester duration. The current project was completed within academic constraints but would benefit from an extended timeline or dedicated project course spanning multiple semesters. The school may consider establishing a "Capstone Project" track specifically for software development initiatives, providing students with more realistic timeframes for building production-ready applications. Additionally, access to development resources such as cloud hosting credits, domain registration, and API keys for educational services would significantly enhance the quality and deployability of student projects.

To the Partners (Owners of the Business Targeted) about the App Developed:

Ubumenyi Hub represents a minimum viable product (MVP) that demonstrates the potential of localized e-learning in Rwanda. We recommend partnering with the Rwanda Education Board and selected pilot schools to test the platform with real students and teachers. This user testing will provide invaluable feedback for prioritizing feature development. The platform requires investment in content creation, particularly video lessons in Kinyarwanda and curriculum-aligned materials. Technical investment in backend infrastructure, content delivery networks (CDN), and mobile optimization will be necessary for scaling beyond the current prototype stage. We welcome partnership discussions for funding, content collaboration, and technical mentorship.

To the Users about Training:

Users are encouraged to explore all features of the platform systematically, starting with the Home page and progressing through Videos, Library, Simulations, and the Dashboard. New users should begin by creating an account to access personalized features including reading progress tracking and forum participation. For the simulation engine, we recommend starting with the community-saved examples before creating custom experiments. Video lessons can be viewed without registration, but creating an account enables bookmarking and progress tracking. Teachers interested in using the platform for their classes should contact the development team for training on quiz creation, forum moderation, and live session planning. User feedback is highly valued and can be submitted through the About Us page or GitHub repository issues.

To the Beneficiaries (Students and the Community) about the New Technological Tool:

Ubumenyi Hub is designed to complement, not replace, traditional classroom learning. Students are encouraged to use the video lessons for pre-class preparation and post-class review, maximizing the effectiveness of face-to-face teacher interactions. The physics simulation engine provides safe, unlimited experimentation opportunities that would be impossible in traditional laboratories — use this feature to explore "what if" scenarios and develop intuitive understanding before formal mathematical treatment. The digital library supports deep reading and note-taking practices that improve comprehension and retention.

Community members are invited to contribute to the discussion forums, share knowledge, and help build a supportive learning ecosystem. As the platform evolves, users can expect additional content in Kinyarwanda, more simulation categories, and enhanced mobile accessibility. Your feedback and participation are essential to making Ubumenyi Hub a truly community-driven educational resource.

APPENDIX

A.1 GitHub Repository Information

The complete source code for the Ubumenyi Hub project is publicly available on GitHub for educational purposes, collaborative development, and version control.

Repository URL: <https://github.com/djmp110/Ubumenyi-HUB>

Repository Name: Ubumenyi-HUB

Owner: djmp110 (MUTUYIMANA Pacifique)

Visibility: Public

Primary Language: HTML (40.4%), CSS (31.2%), JavaScript (28.4%)

Total Commits: 17

Repository Contents:

- **index.html** — Home page and platform hub
- **signup.html** — User registration interface
- **signin.html** — User authentication interface
- **videos.html** — Video lesson library
- **library.html** — Digital reader and personal library
- **simulation.html** — Interactive physics simulation engine
- **online.html** — Dashboard with forums and quizzes
- **about.html** — Platform information and contact
- **admin.html** — Administrative interface
- **style/** — CSS stylesheets directory
- **js/** — JavaScript modules directory
- **image/** — Media assets directory
- **database/** — Data files and JSON schemas
- **.vscode/** — Development environment configuration

The repository is actively maintained and welcomes contributions from the developer community. Issues, feature requests, and pull requests can be submitted through the GitHub interface.

A.2 Project Metadata

Attribute	Value
Project Name	Ubumenyi Hub
Project Type	Web-Based E-Learning Platform
Developer	MUTUYIMANA Pacifique
Registration Number	225031462
Institution	University of Rwanda, College of Science and Technology
Department	Computer and Software Engineering
Module	Web Design
Academic Year	2025-2026
Development Period	2025-2026 Academic Year
Deployment Status	Live on GitHub Pages
License	Open Source (Educational Use)

A.3 Acknowledgments

This project was developed as part of the Web Design module assessment at the University of Rwanda, College of Science and Technology, School of ICT. The author extends gratitude to the faculty members of the Computer and Software Engineering Department for their guidance, technical mentorship, and support throughout the development process.

Special appreciation goes to the open-source community for providing the foundational web technologies (HTML5, CSS3, JavaScript) and educational resources that made this project possible. The YouTube platform and its embed API are acknowledged for enabling the video lesson delivery component.

The author thanks fellow students and peers who provided feedback during testing phases and contributed to the refinement of the user interface and experience. The name "Ubumenyi" celebrates Rwanda's linguistic heritage and the universal human pursuit of knowledge.